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1 F:\Python\Training_Voice\.venv\Scripts\python.exe F:\Python\
  Training_Voice\training_voice.py
2 GPU: NVIDIA GeForce RTX 4060 Laptop GPU
3 GPU Memory: 8.0 GB
4 GPU Memory Allocated: 0.00 GB
5 2025-07-10 19:29:20,706 - INFO - Configuration: {'csv_file_path': 'F:/
  Python/Training_Voice/unified_dataset.csv', 'batch_size': 32, '
  learning_rate': 0.001, 'num_epochs': 100, 'patience': 25, 'weight_decay
  ': 1e-05, 'use_mixed_precision': True, 'label_smoothing': 0.1, '
  hidden_size': 256, 'dropout': 0.3}
6 2025-07-10 19:29:20,706 - INFO - Loading data...
7 2025-07-10 19:29:21,056 - INFO - Dataset shape: (51937, 42)
8 2025-07-10 19:29:21,057 - INFO - Features shape: (51937, 40)
9 2025-07-10 19:29:21,072 - INFO - Unique emotions: ['angry' 'disgust' '
  fearful' 'happy' 'neutral' 'sad' 'surprised']
10 2025-07-10 19:29:21,076 - INFO - Class distribution: {'angry': 41762, '
  disgust': 2071, 'fearful': 2071, 'happy': 2071, 'neutral': 1887, 'sad': 2071
  , 'surprised': 4}
11 2025-07-10 19:29:21,108 - WARNING - Found extreme values after
  normalization. Clipping to [-10, 10]
12 2025-07-10 19:29:21,114 - INFO - Feature statistics after normalization:
13 2025-07-10 19:29:21,116 - INFO - Mean: 0.000004
14 2025-07-10 19:29:21,125 - INFO - Std: 0.999948
15 2025-07-10 19:29:21,126 - INFO - Min: -10.000000
16 2025-07-10 19:29:21,127 - INFO - Max: 9.512881
17 2025-07-10 19:29:21,165 - INFO - Data splits:
18 2025-07-10 19:29:21,165 - INFO - Train: 36355 samples (70.0%)
19 2025-07-10 19:29:21,165 - INFO - Validation: 5194 samples (10.0%)
20 2025-07-10 19:29:21,165 - INFO - Test: 10388 samples (20.0%)
21 2025-07-10 19:29:21,175 - INFO - Model initialized with 119,303
  parameters
22 2025-07-10 19:29:22,346 - INFO - Starting training...
23 2025-07-10 19:29:28,593 - INFO - Epoch 1/100 (6.25s)
24 2025-07-10 19:29:28,594 - INFO - Train - Loss: 0.8526, Acc: 0.8439, F1:
  0.8341
25 2025-07-10 19:29:28,594 - INFO - Val - Loss: 0.6874, Acc: 0.9093, F1: 0
  .9046
26 2025-07-10 19:29:28,594 - INFO - LR: 1.00e-03
27 2025-07-10 19:29:28,599 - INFO - New best F1 score: 0.9046
28 2025-07-10 19:29:34,477 - INFO - Epoch 2/100 (5.81s)
29 2025-07-10 19:29:34,478 - INFO - Train - Loss: 0.7435, Acc: 0.8882, F1:
  0.8815
30 2025-07-10 19:29:34,478 - INFO - Val - Loss: 0.6761, Acc: 0.9087, F1: 0
  .9054
31 2025-07-10 19:29:34,478 - INFO - LR: 1.00e-03
32 2025-07-10 19:29:34,481 - INFO - New best F1 score: 0.9054
33 2025-07-10 19:29:40,178 - INFO - Epoch 3/100 (5.62s)
34 2025-07-10 19:29:40,179 - INFO - Train - Loss: 0.7269, Acc: 0.8924, F1:
  0.8863
35 2025-07-10 19:29:40,179 - INFO - Val - Loss: 0.6699, Acc: 0.9116, F1: 0
  .9039
36 2025-07-10 19:29:40,179 - INFO - LR: 1.00e-03
37 2025-07-10 19:29:46,082 - INFO - Epoch 4/100 (5.84s)
38 2025-07-10 19:29:46,082 - INFO - Train - Loss: 0.7155, Acc: 0.8962, F1:
  0.8908
39 2025-07-10 19:29:46,082 - INFO - Val - Loss: 0.6698, Acc: 0.9130, F1: 0
  .9095
40 2025-07-10 19:29:46,082 - INFO - LR: 1.00e-03
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41 2025-07-10 19:29:46,086 - INFO - New best F1 score: 0.9095
42 2025-07-10 19:29:51,964 - INFO - Epoch 5/100 (5.81s)
43 2025-07-10 19:29:51,964 - INFO - Train - Loss: 0.7108, Acc: 0.8985, F1: 0.8934
44 2025-07-10 19:29:51,964 - INFO - Val - Loss: 0.6641, Acc: 0.9176, F1: 0.9141
45 2025-07-10 19:29:51,964 - INFO - LR: 1.00e-03
46 2025-07-10 19:29:51,968 - INFO - New best F1 score: 0.9141
47 2025-07-10 19:29:58,297 - INFO - Epoch 6/100 (6.25s)
48 2025-07-10 19:29:58,297 - INFO - Train - Loss: 0.7048, Acc: 0.9007, F1: 0.8963
49 2025-07-10 19:29:58,297 - INFO - Val - Loss: 0.6644, Acc: 0.9134, F1: 0.9088
50 2025-07-10 19:29:58,297 - INFO - LR: 1.00e-03
51 2025-07-10 19:30:04,793 - INFO - Epoch 7/100 (6.43s)
52 2025-07-10 19:30:04,794 - INFO - Train - Loss: 0.7023, Acc: 0.9024, F1: 0.8982
53 2025-07-10 19:30:04,794 - INFO - Val - Loss: 0.6618, Acc: 0.9151, F1: 0.9094
54 2025-07-10 19:30:04,794 - INFO - LR: 1.00e-03
55 2025-07-10 19:30:11,540 - INFO - Epoch 8/100 (6.68s)
56 2025-07-10 19:30:11,540 - INFO - Train - Loss: 0.7000, Acc: 0.9025, F1: 0.8984
57 2025-07-10 19:30:11,540 - INFO - Val - Loss: 0.6600, Acc: 0.9182, F1: 0.9143
58 2025-07-10 19:30:11,541 - INFO - LR: 1.00e-03
59 2025-07-10 19:30:11,545 - INFO - New best F1 score: 0.9143
60 2025-07-10 19:30:17,971 - INFO - Epoch 9/100 (6.35s)
61 2025-07-10 19:30:17,971 - INFO - Train - Loss: 0.6948, Acc: 0.9061, F1: 0.9025
62 2025-07-10 19:30:17,971 - INFO - Val - Loss: 0.6592, Acc: 0.9126, F1: 0.9083
63 2025-07-10 19:30:17,971 - INFO - LR: 1.00e-03
64 2025-07-10 19:30:24,370 - INFO - Epoch 10/100 (6.32s)
65 2025-07-10 19:30:24,370 - INFO - Train - Loss: 0.6946, Acc: 0.9043, F1: 0.9004
66 2025-07-10 19:30:24,370 - INFO - Val - Loss: 0.6547, Acc: 0.9226, F1: 0.9212
67 2025-07-10 19:30:24,370 - INFO - LR: 1.00e-03
68 2025-07-10 19:30:24,374 - INFO - New best F1 score: 0.9212
69 2025-07-10 19:30:30,067 - INFO - Epoch 11/100 (5.62s)
70 2025-07-10 19:30:30,067 - INFO - Train - Loss: 0.6918, Acc: 0.9069, F1: 0.9031
71 2025-07-10 19:30:30,067 - INFO - Val - Loss: 0.6560, Acc: 0.9205, F1: 0.9167
72 2025-07-10 19:30:30,067 - INFO - LR: 1.00e-03
73 2025-07-10 19:30:35,782 - INFO - Epoch 12/100 (5.65s)
74 2025-07-10 19:30:35,782 - INFO - Train - Loss: 0.6913, Acc: 0.9072, F1: 0.9036
75 2025-07-10 19:30:35,782 - INFO - Val - Loss: 0.6565, Acc: 0.9195, F1: 0.9183
76 2025-07-10 19:30:35,782 - INFO - LR: 1.00e-03
77 2025-07-10 19:30:41,502 - INFO - Epoch 13/100 (5.65s)
78 2025-07-10 19:30:41,502 - INFO - Train - Loss: 0.6888, Acc: 0.9073, F1: 0.9036
79 2025-07-10 19:30:41,502 - INFO - Val - Loss: 0.6521, Acc: 0.9214, F1: 0.9195
80 2025-07-10 19:30:41,502 - INFO - LR: 1.00e-03

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81 2025-07-10 19:30:47,283 - INFO - Epoch 14/100 (5.71s)
82 2025-07-10 19:30:47,283 - INFO - Train - Loss: 0.6891, Acc: 0.9078, F1
   : 0.9041
83 2025-07-10 19:30:47,283 - INFO - Val  - Loss: 0.6521, Acc: 0.9218, F1:
   0.9194
84 2025-07-10 19:30:47,283 - INFO - LR: 1.00e-03
85 2025-07-10 19:30:52,996 - INFO - Epoch 15/100 (5.64s)
86 2025-07-10 19:30:52,996 - INFO - Train - Loss: 0.6871, Acc: 0.9091, F1
   : 0.9060
87 2025-07-10 19:30:52,996 - INFO - Val  - Loss: 0.6495, Acc: 0.9211, F1:
   0.9176
88 2025-07-10 19:30:52,996 - INFO - LR: 1.00e-03
89 2025-07-10 19:30:58,772 - INFO - Epoch 16/100 (5.70s)
90 2025-07-10 19:30:58,773 - INFO - Train - Loss: 0.6849, Acc: 0.9101, F1
   : 0.9073
91 2025-07-10 19:30:58,773 - INFO - Val  - Loss: 0.6494, Acc: 0.9234, F1:
   0.9219
92 2025-07-10 19:30:58,773 - INFO - LR: 1.00e-03
93 2025-07-10 19:30:58,776 - INFO - New best F1 score: 0.9219
94 2025-07-10 19:31:04,479 - INFO - Epoch 17/100 (5.63s)
95 2025-07-10 19:31:04,479 - INFO - Train - Loss: 0.6831, Acc: 0.9121, F1
   : 0.9095
96 2025-07-10 19:31:04,479 - INFO - Val  - Loss: 0.6509, Acc: 0.9199, F1:
   0.9179
97 2025-07-10 19:31:04,479 - INFO - LR: 1.00e-03
98 2025-07-10 19:31:10,237 - INFO - Epoch 18/100 (5.68s)
99 2025-07-10 19:31:10,237 - INFO - Train - Loss: 0.6837, Acc: 0.9110, F1
   : 0.9081
100 2025-07-10 19:31:10,237 - INFO - Val  - Loss: 0.6494, Acc: 0.9245, F1:
    0.9214
101 2025-07-10 19:31:10,237 - INFO - LR: 1.00e-03
102 2025-07-10 19:31:16,064 - INFO - Epoch 19/100 (5.76s)
103 2025-07-10 19:31:16,064 - INFO - Train - Loss: 0.6830, Acc: 0.9124, F1
   : 0.9097
104 2025-07-10 19:31:16,064 - INFO - Val  - Loss: 0.6493, Acc: 0.9220, F1:
   0.9195
105 2025-07-10 19:31:16,064 - INFO - LR: 1.00e-03
106 2025-07-10 19:31:21,810 - INFO - Epoch 20/100 (5.67s)
107 2025-07-10 19:31:21,811 - INFO - Train - Loss: 0.6817, Acc: 0.9124, F1
   : 0.9096
108 2025-07-10 19:31:21,811 - INFO - Val  - Loss: 0.6482, Acc: 0.9240, F1:
   0.9217
109 2025-07-10 19:31:21,811 - INFO - LR: 1.00e-03
110 2025-07-10 19:31:27,717 - INFO - Epoch 21/100 (5.83s)
111 2025-07-10 19:31:27,717 - INFO - Train - Loss: 0.6807, Acc: 0.9129, F1
   : 0.9105
112 2025-07-10 19:31:27,717 - INFO - Val  - Loss: 0.6477, Acc: 0.9238, F1:
   0.9204
113 2025-07-10 19:31:27,717 - INFO - LR: 1.00e-03
114 2025-07-10 19:31:33,567 - INFO - Epoch 22/100 (5.78s)
115 2025-07-10 19:31:33,567 - INFO - Train - Loss: 0.6812, Acc: 0.9126, F1
   : 0.9097
116 2025-07-10 19:31:33,567 - INFO - Val  - Loss: 0.6466, Acc: 0.9249, F1:
   0.9236
117 2025-07-10 19:31:33,567 - INFO - LR: 1.00e-03
118 2025-07-10 19:31:33,571 - INFO - New best F1 score: 0.9236
119 2025-07-10 19:31:39,574 - INFO - Epoch 23/100 (5.93s)
120 2025-07-10 19:31:39,574 - INFO - Train - Loss: 0.6777, Acc: 0.9143, F1

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120 : 0.9117
121 2025-07-10 19:31:39,574 - INFO - Val - Loss: 0.6467, Acc: 0.9213, F1: 0.9182
122 2025-07-10 19:31:39,574 - INFO - LR: 1.00e-03
123 2025-07-10 19:31:45,335 - INFO - Epoch 24/100 (5.68s)
124 2025-07-10 19:31:45,335 - INFO - Train - Loss: 0.6778, Acc: 0.9142, F1 : 0.9119
125 2025-07-10 19:31:45,335 - INFO - Val - Loss: 0.6471, Acc: 0.9218, F1: 0.9211
126 2025-07-10 19:31:45,335 - INFO - LR: 1.00e-03
127 2025-07-10 19:31:51,149 - INFO - Epoch 25/100 (5.74s)
128 2025-07-10 19:31:51,149 - INFO - Train - Loss: 0.6761, Acc: 0.9149, F1 : 0.9127
129 2025-07-10 19:31:51,149 - INFO - Val - Loss: 0.6438, Acc: 0.9255, F1: 0.9238
130 2025-07-10 19:31:51,149 - INFO - LR: 1.00e-03
131 2025-07-10 19:31:51,154 - INFO - New best F1 score: 0.9238
132 2025-07-10 19:31:57,304 - INFO - Epoch 26/100 (6.06s)
133 2025-07-10 19:31:57,304 - INFO - Train - Loss: 0.6788, Acc: 0.9135, F1 : 0.9114
134 2025-07-10 19:31:57,304 - INFO - Val - Loss: 0.6451, Acc: 0.9266, F1: 0.9232
135 2025-07-10 19:31:57,304 - INFO - LR: 1.00e-03
136 2025-07-10 19:32:03,413 - INFO - Epoch 27/100 (6.04s)
137 2025-07-10 19:32:03,413 - INFO - Train - Loss: 0.6767, Acc: 0.9156, F1 : 0.9132
138 2025-07-10 19:32:03,413 - INFO - Val - Loss: 0.6428, Acc: 0.9274, F1: 0.9255
139 2025-07-10 19:32:03,413 - INFO - LR: 1.00e-03
140 2025-07-10 19:32:03,416 - INFO - New best F1 score: 0.9255
141 2025-07-10 19:32:09,773 - INFO - Epoch 28/100 (6.29s)
142 2025-07-10 19:32:09,773 - INFO - Train - Loss: 0.6742, Acc: 0.9175, F1 : 0.9155
143 2025-07-10 19:32:09,773 - INFO - Val - Loss: 0.6433, Acc: 0.9251, F1: 0.9225
144 2025-07-10 19:32:09,773 - INFO - LR: 1.00e-03
145 2025-07-10 19:32:16,300 - INFO - Epoch 29/100 (6.46s)
146 2025-07-10 19:32:16,300 - INFO - Train - Loss: 0.6751, Acc: 0.9163, F1 : 0.9140
147 2025-07-10 19:32:16,300 - INFO - Val - Loss: 0.6452, Acc: 0.9255, F1: 0.9235
148 2025-07-10 19:32:16,300 - INFO - LR: 1.00e-03
149 2025-07-10 19:32:22,669 - INFO - Epoch 30/100 (6.30s)
150 2025-07-10 19:32:22,669 - INFO - Train - Loss: 0.6761, Acc: 0.9151, F1 : 0.9129
151 2025-07-10 19:32:22,669 - INFO - Val - Loss: 0.6455, Acc: 0.9232, F1: 0.9189
152 2025-07-10 19:32:22,669 - INFO - LR: 1.00e-03
153 2025-07-10 19:32:28,471 - INFO - Epoch 31/100 (5.73s)
154 2025-07-10 19:32:28,471 - INFO - Train - Loss: 0.6738, Acc: 0.9166, F1 : 0.9143
155 2025-07-10 19:32:28,471 - INFO - Val - Loss: 0.6426, Acc: 0.9257, F1: 0.9232
156 2025-07-10 19:32:28,472 - INFO - LR: 1.00e-03
157 2025-07-10 19:32:34,208 - INFO - Epoch 32/100 (5.66s)
158 2025-07-10 19:32:34,208 - INFO - Train - Loss: 0.6739, Acc: 0.9167, F1 : 0.9147
159 2025-07-10 19:32:34,208 - INFO - Val - Loss: 0.6437, Acc: 0.9247, F1:

159 0.9229
160 2025-07-10 19:32:34,208 - INFO - LR: 1.00e-03
161 2025-07-10 19:32:39,874 - INFO - Epoch 33/100 (5.60s)
162 2025-07-10 19:32:39,874 - INFO - Train - Loss: 0.6738, Acc: 0.9162, F1 : 0.9140
163 2025-07-10 19:32:39,874 - INFO - Val - Loss: 0.6450, Acc: 0.9241, F1: 0.9226
164 2025-07-10 19:32:39,874 - INFO - LR: 1.00e-03
165 2025-07-10 19:32:45,562 - INFO - Epoch 34/100 (5.62s)
166 2025-07-10 19:32:45,562 - INFO - Train - Loss: 0.6734, Acc: 0.9166, F1 : 0.9142
167 2025-07-10 19:32:45,562 - INFO - Val - Loss: 0.6427, Acc: 0.9266, F1: 0.9250
168 2025-07-10 19:32:45,562 - INFO - LR: 1.00e-03
169 2025-07-10 19:32:51,261 - INFO - Epoch 35/100 (5.62s)
170 2025-07-10 19:32:51,261 - INFO - Train - Loss: 0.6714, Acc: 0.9182, F1 : 0.9161
171 2025-07-10 19:32:51,261 - INFO - Val - Loss: 0.6423, Acc: 0.9270, F1: 0.9247
172 2025-07-10 19:32:51,261 - INFO - LR: 1.00e-03
173 2025-07-10 19:32:57,018 - INFO - Epoch 36/100 (5.68s)
174 2025-07-10 19:32:57,018 - INFO - Train - Loss: 0.6722, Acc: 0.9180, F1 : 0.9161
175 2025-07-10 19:32:57,018 - INFO - Val - Loss: 0.6425, Acc: 0.9288, F1: 0.9283
176 2025-07-10 19:32:57,018 - INFO - LR: 1.00e-03
177 2025-07-10 19:32:57,021 - INFO - New best F1 score: 0.9283
178 2025-07-10 19:33:02,700 - INFO - Epoch 37/100 (5.61s)
179 2025-07-10 19:33:02,700 - INFO - Train - Loss: 0.6696, Acc: 0.9184, F1 : 0.9165
180 2025-07-10 19:33:02,700 - INFO - Val - Loss: 0.6427, Acc: 0.9266, F1: 0.9250
181 2025-07-10 19:33:02,700 - INFO - LR: 1.00e-03
182 2025-07-10 19:33:08,862 - INFO - Epoch 38/100 (6.09s)
183 2025-07-10 19:33:08,862 - INFO - Train - Loss: 0.6720, Acc: 0.9176, F1 : 0.9154
184 2025-07-10 19:33:08,863 - INFO - Val - Loss: 0.6407, Acc: 0.9284, F1: 0.9271
185 2025-07-10 19:33:08,863 - INFO - LR: 1.00e-03
186 2025-07-10 19:33:15,627 - INFO - Epoch 39/100 (6.68s)
187 2025-07-10 19:33:15,627 - INFO - Train - Loss: 0.6689, Acc: 0.9188, F1 : 0.9170
188 2025-07-10 19:33:15,627 - INFO - Val - Loss: 0.6412, Acc: 0.9263, F1: 0.9259
189 2025-07-10 19:33:15,627 - INFO - LR: 1.00e-03
190 2025-07-10 19:33:22,134 - INFO - Epoch 40/100 (6.42s)
191 2025-07-10 19:33:22,134 - INFO - Train - Loss: 0.6705, Acc: 0.9189, F1 : 0.9171
192 2025-07-10 19:33:22,134 - INFO - Val - Loss: 0.6398, Acc: 0.9266, F1: 0.9261
193 2025-07-10 19:33:22,134 - INFO - LR: 1.00e-03
194 2025-07-10 19:33:29,228 - INFO - Epoch 41/100 (7.00s)
195 2025-07-10 19:33:29,228 - INFO - Train - Loss: 0.6709, Acc: 0.9188, F1 : 0.9170
196 2025-07-10 19:33:29,228 - INFO - Val - Loss: 0.6423, Acc: 0.9290, F1: 0.9275
197 2025-07-10 19:33:29,228 - INFO - LR: 1.00e-03
198 2025-07-10 19:33:35,774 - INFO - Epoch 42/100 (6.48s)

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199 2025-07-10 19:33:35,774 - INFO - Train - Loss: 0.6688, Acc: 0.9202, F1
    : 0.9182
200 2025-07-10 19:33:35,774 - INFO - Val  - Loss: 0.6422, Acc: 0.9255, F1:
    0.9245
201 2025-07-10 19:33:35,774 - INFO - LR: 1.00e-03
202 2025-07-10 19:33:42,231 - INFO - Epoch 43/100 (6.39s)
203 2025-07-10 19:33:42,231 - INFO - Train - Loss: 0.6688, Acc: 0.9188, F1
    : 0.9170
204 2025-07-10 19:33:42,231 - INFO - Val  - Loss: 0.6400, Acc: 0.9301, F1:
    0.9286
205 2025-07-10 19:33:42,231 - INFO - LR: 1.00e-03
206 2025-07-10 19:33:42,235 - INFO - New best F1 score: 0.9286
207 2025-07-10 19:33:49,188 - INFO - Epoch 44/100 (6.88s)
208 2025-07-10 19:33:49,189 - INFO - Train - Loss: 0.6680, Acc: 0.9203, F1
    : 0.9186
209 2025-07-10 19:33:49,189 - INFO - Val  - Loss: 0.6408, Acc: 0.9265, F1:
    0.9250
210 2025-07-10 19:33:49,189 - INFO - LR: 1.00e-03
211 2025-07-10 19:33:56,298 - INFO - Epoch 45/100 (7.03s)
212 2025-07-10 19:33:56,298 - INFO - Train - Loss: 0.6666, Acc: 0.9199, F1
    : 0.9181
213 2025-07-10 19:33:56,298 - INFO - Val  - Loss: 0.6397, Acc: 0.9282, F1:
    0.9271
214 2025-07-10 19:33:56,298 - INFO - LR: 1.00e-03
215 2025-07-10 19:34:03,591 - INFO - Epoch 46/100 (7.22s)
216 2025-07-10 19:34:03,591 - INFO - Train - Loss: 0.6663, Acc: 0.9215, F1
    : 0.9199
217 2025-07-10 19:34:03,591 - INFO - Val  - Loss: 0.6409, Acc: 0.9265, F1:
    0.9259
218 2025-07-10 19:34:03,591 - INFO - LR: 1.00e-03
219 2025-07-10 19:34:09,767 - INFO - Epoch 47/100 (6.10s)
220 2025-07-10 19:34:09,767 - INFO - Train - Loss: 0.6655, Acc: 0.9209, F1
    : 0.9191
221 2025-07-10 19:34:09,768 - INFO - Val  - Loss: 0.6407, Acc: 0.9274, F1:
    0.9252
222 2025-07-10 19:34:09,768 - INFO - LR: 1.00e-03
223 2025-07-10 19:34:16,010 - INFO - Epoch 48/100 (6.17s)
224 2025-07-10 19:34:16,010 - INFO - Train - Loss: 0.6646, Acc: 0.9215, F1
    : 0.9199
225 2025-07-10 19:34:16,010 - INFO - Val  - Loss: 0.6424, Acc: 0.9276, F1:
    0.9245
226 2025-07-10 19:34:16,010 - INFO - LR: 1.00e-03
227 2025-07-10 19:34:22,447 - INFO - Epoch 49/100 (6.37s)
228 2025-07-10 19:34:22,447 - INFO - Train - Loss: 0.6666, Acc: 0.9205, F1
    : 0.9190
229 2025-07-10 19:34:22,447 - INFO - Val  - Loss: 0.6411, Acc: 0.9274, F1:
    0.9254
230 2025-07-10 19:34:22,447 - INFO - LR: 1.00e-03
231 2025-07-10 19:34:28,905 - INFO - Epoch 50/100 (6.39s)
232 2025-07-10 19:34:28,905 - INFO - Train - Loss: 0.6662, Acc: 0.9214, F1
    : 0.9197
233 2025-07-10 19:34:28,905 - INFO - Val  - Loss: 0.6406, Acc: 0.9293, F1:
    0.9281
234 2025-07-10 19:34:28,905 - INFO - LR: 1.00e-03
235 2025-07-10 19:34:35,184 - INFO - Epoch 51/100 (6.21s)
236 2025-07-10 19:34:35,184 - INFO - Train - Loss: 0.6646, Acc: 0.9210, F1
    : 0.9192
237 2025-07-10 19:34:35,185 - INFO - Val  - Loss: 0.6424, Acc: 0.9282, F1:

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237 0.9257
238 2025-07-10 19:34:35,185 - INFO - LR: 1.00e-03
239 2025-07-10 19:34:41,548 - INFO - Epoch 52/100 (6.28s)
240 2025-07-10 19:34:41,549 - INFO - Train - Loss: 0.6657, Acc: 0.9202, F1: 0.9185
241 2025-07-10 19:34:41,549 - INFO - Val - Loss: 0.6409, Acc: 0.9274, F1: 0.9254
242 2025-07-10 19:34:41,549 - INFO - LR: 1.00e-03
243 2025-07-10 19:34:47,967 - INFO - Epoch 53/100 (6.35s)
244 2025-07-10 19:34:47,967 - INFO - Train - Loss: 0.6652, Acc: 0.9223, F1: 0.9206
245 2025-07-10 19:34:47,967 - INFO - Val - Loss: 0.6407, Acc: 0.9293, F1: 0.9279
246 2025-07-10 19:34:47,967 - INFO - LR: 1.00e-03
247 2025-07-10 19:34:54,236 - INFO - Epoch 54/100 (6.19s)
248 2025-07-10 19:34:54,236 - INFO - Train - Loss: 0.6654, Acc: 0.9216, F1: 0.9199
249 2025-07-10 19:34:54,236 - INFO - Val - Loss: 0.6382, Acc: 0.9282, F1: 0.9271
250 2025-07-10 19:34:54,236 - INFO - LR: 5.00e-04
251 2025-07-10 19:35:00,662 - INFO - Epoch 55/100 (6.35s)
252 2025-07-10 19:35:00,662 - INFO - Train - Loss: 0.6602, Acc: 0.9238, F1: 0.9224
253 2025-07-10 19:35:00,662 - INFO - Val - Loss: 0.6392, Acc: 0.9297, F1: 0.9273
254 2025-07-10 19:35:00,662 - INFO - LR: 5.00e-04
255 2025-07-10 19:35:06,839 - INFO - Epoch 56/100 (6.11s)
256 2025-07-10 19:35:06,839 - INFO - Train - Loss: 0.6607, Acc: 0.9255, F1: 0.9242
257 2025-07-10 19:35:06,839 - INFO - Val - Loss: 0.6383, Acc: 0.9295, F1: 0.9277
258 2025-07-10 19:35:06,839 - INFO - LR: 5.00e-04
259 2025-07-10 19:35:28,214 - INFO - Epoch 57/100 (21.31s)
260 2025-07-10 19:35:28,215 - INFO - Train - Loss: 0.6577, Acc: 0.9249, F1: 0.9236
261 2025-07-10 19:35:28,215 - INFO - Val - Loss: 0.6388, Acc: 0.9295, F1: 0.9278
262 2025-07-10 19:35:28,215 - INFO - LR: 5.00e-04
263 2025-07-10 19:35:48,688 - INFO - Epoch 58/100 (20.22s)
264 2025-07-10 19:35:48,688 - INFO - Train - Loss: 0.6602, Acc: 0.9239, F1: 0.9224
265 2025-07-10 19:35:48,688 - INFO - Val - Loss: 0.6390, Acc: 0.9272, F1: 0.9271
266 2025-07-10 19:35:48,688 - INFO - LR: 5.00e-04
267 2025-07-10 19:35:54,523 - INFO - Epoch 59/100 (5.76s)
268 2025-07-10 19:35:54,523 - INFO - Train - Loss: 0.6565, Acc: 0.9261, F1: 0.9249
269 2025-07-10 19:35:54,523 - INFO - Val - Loss: 0.6382, Acc: 0.9261, F1: 0.9254
270 2025-07-10 19:35:54,523 - INFO - LR: 5.00e-04
271 2025-07-10 19:36:00,828 - INFO - Epoch 60/100 (6.23s)
272 2025-07-10 19:36:00,828 - INFO - Train - Loss: 0.6572, Acc: 0.9266, F1: 0.9253
273 2025-07-10 19:36:00,828 - INFO - Val - Loss: 0.6392, Acc: 0.9280, F1: 0.9277
274 2025-07-10 19:36:00,829 - INFO - LR: 5.00e-04
275 2025-07-10 19:36:07,512 - INFO - Epoch 61/100 (6.61s)
276 2025-07-10 19:36:07,512 - INFO - Train - Loss: 0.6565, Acc: 0.9267, F1:

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276 : 0.9255
277 2025-07-10 19:36:07,512 - INFO - Val - Loss: 0.6389, Acc: 0.9274, F1:
    0.9266
278 2025-07-10 19:36:07,512 - INFO - LR: 5.00e-04
279 2025-07-10 19:36:07,513 - INFO - Early stopping at epoch 61
280 2025-07-10 19:36:07,526 - INFO - Loaded best model weights
281 2025-07-10 19:36:09,598 - INFO - Metrics plot saved to
    training_metrics.png
282 2025-07-10 19:36:09,598 - INFO - Testing final model...
283 Testing: 100%|██████████| 325/325 [00:00<00:00, 883.45it/s]
284 2025-07-10 19:36:10,002 - INFO -
285 Training completed in 407.66 seconds
286 2025-07-10 19:36:10,002 - INFO - Best validation F1: 0.9286
287 2025-07-10 19:36:10,002 - INFO - Test accuracy: 0.9259
288 2025-07-10 19:36:10,002 - INFO - Test F1 score: 0.9248
289 2025-07-10 19:36:10,002 - INFO -
290 Classification Report:
291      precision    recall  f1-score   support
292
293     angry       0.99      0.99      0.99     8353
294     disgust     0.75      0.56      0.64      414
295     fearful     0.81      0.52      0.63      414
296     happy       0.75      0.62      0.68      414
297     neutral     0.55      0.78      0.65      378
298     sad         0.62      0.82      0.70      414
299     surprised    0.00      0.00      0.00         1
300
301     accuracy                0.93    10388
302     macro avg       0.64      0.61      0.61    10388
303     weighted avg     0.93      0.93      0.92    10388
304
305 2025-07-10 19:36:10,748 - INFO - Confusion matrix saved to
    confusion_matrix.png
306 2025-07-10 19:36:10,754 - INFO - Model saved to best_model.pth
307 2025-07-10 19:36:10,754 - INFO - Preprocessing artifacts saved to
    preprocessing_artifacts.pkl
308 2025-07-10 19:36:10,754 - INFO - Training completed successfully!
309
310 Process finished with exit code 0
311

```